

CS 505 Module Four Activity: Mathematical Expressions Template

Complete this template by replacing the bracketed text with the relevant information.

1. Represent R as an adjacency matrix for the information provided:

Set $A = \{a, b, c, d\}$ and the relation $R = \{(a, b), (b, c), (c, d), (a, c)\}$

—	a	b	c	d
a	0	1	1	0
b	0	0	1	0
c	0	0	0	1
d	0	0	0	0

2. Square the adjacency matrix to find which pairs of elements are connected by a path of length 2 for the information provided:

Set $A = \{a, b, c\}$ and a relation R on A defined by $R = \{(a, a), (a, b), (b, b), (b, c), (c, c)\}$

Adjacency Matrix – B

—	a	b	c
a	1	1	0
b	0	1	1
c	0	0	1

Square of Adjacency = B^2

$$B^2 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

The pairs connected by a path of length 2 are:

$(a, a), (a, b), (a, c), (b, b), (b, c), (c, c)$

Because those entries in B^2 are nonzero.

3. Perform the matrix calculations for the information provided:

- A. Calculate $A + B$, $A - B$, and $A \times B$ for the information provided:

Matrices $A = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$

$A + B$

$$\begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2+1 & 1+4 \\ 3+0 & 0+2 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 3 & 2 \end{bmatrix}$$

$A - B$

$$\begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2-1 & 1-4 \\ 3-0 & 0-2 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 3 & -2 \end{bmatrix}$$

$A \times B$

$$\begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2+0 & 8+2 \\ 3+0 & 12+0 \end{bmatrix} = \begin{bmatrix} 2 & 10 \\ 3 & 12 \end{bmatrix}$$

- B. Calculate $A + B$, $A - B$, and $A \times B$ for the information provided:

Matrices $A = [2 \ 1; 0 \ 2]$ and $B = [3 \ 0; 1 \ 4]$

$A + B$

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 2+3 & 1+0 \\ 0+1 & 2+4 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 6 \end{bmatrix}$$

$A - B$

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 2-3 & 1-0 \\ 0-1 & 2-4 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & -2 \end{bmatrix}$$

$A \times B$

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6+1 & 0+4 \\ 0+2 & 0+8 \end{bmatrix} = \begin{bmatrix} 7 & 4 \\ 2 & 8 \end{bmatrix}$$

4. Compute $A \times B$ for the information provided:

$A = [2 \ 1; -4 \ 2]$ $B = [5 \ -6 \ 8; 2 \ 1 \ 0]$ $C = [2 \ 1; 0 \ 3; 1 \ 6]$

$$\begin{bmatrix} 2 & 1 \\ -4 & 2 \end{bmatrix} \times \begin{bmatrix} 5 & -6 & 8 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 10+2 & -12+1 & 16+0 \\ -20+4 & 24+2 & -32+0 \end{bmatrix} = \begin{bmatrix} 12 & -11 & 16 \\ -16 & 26 & -32 \end{bmatrix}$$

5. Compute $C \times A$ for the information provided:

$A = [2 \ 1; -4 \ 2]$ $B = [5 \ -6 \ 8; 2 \ 1 \ 0]$ $C = [2 \ 1; 0 \ 3; 1 \ 6]$

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 6 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 4-4 & 2+2 \\ 0-12 & 0+6 \\ 2-24 & 1+12 \end{bmatrix} = \begin{bmatrix} 0 & 4 \\ -12 & 6 \\ -22 & 13 \end{bmatrix}$$

6. Compute $A \times A$ for the information provided:

$A = [2 \ 1; -4 \ 2]$ $B = [5 \ -6 \ 8; 2 \ 1 \ 0]$ $C = [2 \ 1; 0 \ 3; 1 \ 6]$

$$\begin{bmatrix} 2 & 1 \\ -4 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 4-4 & 2+2 \\ -8-8 & -4+4 \end{bmatrix} = \begin{bmatrix} 0 & 4 \\ -16 & 0 \end{bmatrix}$$

7. Find the dimensions of matrix A for the information provided:

$A = [1 \ 3 \ 0; 5 \ 8 \ 9; -4 \ 2 \ 7]$

Dimensions of $A = 3 \times 3$

8. Find the entries at a_{31} and a_{22} for the information provided:

$A = [1 \ 3 \ 0; 5 \ 8 \ 9; -4 \ 2 \ 7]$

$$a_{31} = -4$$

$$a_{22} = 8$$

9. Find $AB - C$ for the information provided:

$$A = \begin{bmatrix} -20 & 6 & 3 \\ 30 & -1 & -40 \\ 8 & 20 & -2 \end{bmatrix}$$

$$B = \begin{bmatrix} 40 & 18 & -10 \\ -5 & 50 & 16 \\ 2 & -20 & -9 \end{bmatrix}$$

$$C = \begin{bmatrix} -90 & -15 & -80 \\ 10 & -20 & 4 \\ -70 & 15 & -8 \end{bmatrix}$$

$$\begin{aligned} AB &= \begin{bmatrix} -20 & 6 & 3 \\ 30 & -1 & -40 \\ 8 & 20 & -2 \end{bmatrix} \times \begin{bmatrix} 40 & 18 & -10 \\ -5 & 50 & 16 \\ 2 & -20 & -9 \end{bmatrix} \\ &= \begin{bmatrix} -800 - 30 + 6 & -360 + 300 - 60 & 200 + 96 - 27 \\ 1200 + 5 - 80 & 540 - 50 + 800 & -300 - 16 + 360 \\ 320 - 100 - 4 & 144 + 1000 + 40 & -80 + 320 + 18 \end{bmatrix} \\ &= \begin{bmatrix} -824 & -120 & 269 \\ 1125 & 1290 & 44 \\ 216 & 1184 & 258 \end{bmatrix} \\ AB - C &= \begin{bmatrix} -824 & -120 & 269 \\ 1125 & 1290 & 44 \\ 216 & 1184 & 258 \end{bmatrix} - \begin{bmatrix} -90 & -15 & -80 \\ 10 & -20 & 4 \\ -70 & 15 & -8 \end{bmatrix} \\ &= \begin{bmatrix} -824 + 90 & -120 + 15 & 269 + 80 \\ 1125 - 10 & 1290 + 20 & 44 - 4 \\ 216 + 70 & 1184 - 15 & 258 + 8 \end{bmatrix} = \begin{bmatrix} -734 & -105 & 349 \\ 1115 & 1310 & 40 \\ 286 & 1169 & 266 \end{bmatrix} \end{aligned}$$